Blockchain-based Cybersecurity Threat Intelligence Sharing Platform by Integrating Incentive Mechanisms

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Abstract. The threat intelligence market facilitates information sharing in cybersecurity, enabling various organizations to exchange insights about potential cyber threats. However, conventional threat intelligence sharing models, such as those based on federated learning or blockchain, have several limitations, including inconsistent data quality, inadequate incentives, and a lack of transparency. Hence, we propose a blockchain-based cybersecurity intelligence-sharing platform enhanced with an incentive mechanism to address challenges in efficiency, security, transparency, and mutual benefit. Central to the platform is a tripartite game-theoretic pricing contract that dynamically calculates optimal rewards for intelligence submissions, considering submission volume, market demand, and service evaluations. This model ensures a balance between intelligence quality and stakeholder benefit maximization, fostering a positive feedback loop that encourages continuous contributions of high-value intelligence. Our analysis also explores the impact of user scale on the accuracy of intelligence evaluation scores, providing insights into the interplay between user behavior, intelligence valuation, and platform performance. The results demonstrate that the proposed platform effectively incentivizes participation, enriches the intelligence sharing ecosystem, and enhances overall cybersecurity resilience.

Keywords: Cybersecurity Threat Intelligence, Blockchain, Sharing, Incentive Mechanisms

1 Introduction

Cyber threat intelligence (CTI) is a crucial information resource in cybersecurity [1, 2], playing a pivotal role in helping organizations understand the current security landscape, identify potential threats, and enhance their defense mechanisms. As cyber threats continue to escalate, the effective utilization of CTI has become increasingly significant. However, most open-source and readily accessible threat intelligence on the internet tends to be of relatively low value. As a result, threat intelligence markets provide an effective mechanism for promoting information sharing within the cybersecurity community. They facilitate the exchange of intelligence on potential cyber threats among different organizations. By participating in these platforms, enterprises, security agencies, and cybersecurity experts can share the intelligence they have collected and analyzed using various techniques and methodologies.

Nevertheless, many high-value threat intelligence datasets involve sensitive corporate information or have significant threat-level implications. In the absence of proper incentives and regulatory frameworks, organizations are generally reluctant to share such valuable intelligence, leading to the well-known issue of "information silos." Federated learning (FL)-based collaborative threat intelligence sharing enables organizations to jointly design, train, and evaluate machine learning models while maintaining data privacy, enabling collaboration without exposing sensitive information [3, 4]. However, FL requires substantial computational resources, particularly on the client side, as each organization must train its local model. This can be resource-intensive, creating barriers for smaller organizations with limited infrastructure.

With the inherent features of decentralized ledgers, data immutability, and transparency, blockchain technology has demonstrated significant potential across multiple industries, including threat intelligence sharing. Researchers have proposed various blockchain-based methods and frameworks (e.g., [5–8]) to facilitate secure and efficient CTI sharing. By integrating blockchain into threat intelligence sharing frameworks, these approaches address key challenges such as data authenticity, secure access control, and trust mechanisms. As the cybersecurity landscape evolves, blockchain-based threat intelligence markets are expected to play a pivotal role in fostering collaborative security efforts across organizations worldwide.

`Challenges.` Despite efforts to promote CTI sharing, several challenges persist: i) `Lack of standardization`. The absence of a unified standard leads to varying formats and protocols, causing interoperability issues and hindering cross-organizational collaboration. Standardized formats and protocols are essential to streamline data exchange. ii) `Data quality and accuracy`. Unverified or inaccurate intelligence can mislead security teams, wasting resources and undermining defense strategies. Establishing an effective evaluation mechanism is crucial to ensure the reliability and value of shared data. iii) `Lack of incentives for participation`. Organizations often lack the motivation to share intelligence, fearing exposure of vulnerabilities. An incentive mechanism is key to matching intelligence supply and demand, improving quality, enhancing platform credibility, and boosting user engagement, which are vital for the system's sustainability.

Therefore, we developed a cyber threat intelligence (CTI) sharing platform using Hyperledger Fabric as the underlying blockchain infrastructure to ensure shared intelligence's authenticity, consistency, and integrity. By employing permissioned blockchain access rules, the platform verifies the identities of intelligence providers and administrators while preventing data tampering by malicious actors. Smart contracts automate predefined rules, such as rewarding providers with points once their intelligence is validated as accurate and useful, enhancing system credibility, and encouraging broader participation. Additionally, a game-theoretic mathematical model optimizes intelligence sharing among platform administrators, providers, and consumers by calculating optimal pricing in reward points. This model maximizes stakeholder benefits, maintains intelligence quality, and fosters active participation in the ecosystem. The following sections detail the platform's design and implementation, demonstrating its effectiveness in creating a secure, transparent, and incentivized CTI-sharing environment.

The main contributions of this paper are summarized as follows.

- We developed a CTI sharing platform on Hyperledger Fabric, a consortium blockchain. Strict access controls ensure that only verified participants can join, preventing malicious interference and ensuring data integrity.
- We propose a reputation and intelligence price calculation model to guarantee the quality of the intelligence shared by users and the accuracy of the price calculated by the incentive mechanism.
- We conducted two sets of experiments to evaluate the impact of user demand and demand sensitivity on intelligence pricing, as well as the effect of the platform user scale on the accuracy of the comprehensive intelligence evaluation score.

2 Background and Preliminaries

2.1 Blockchain

Fabric [9] is a high-performance, highly scalable enterprise-grade blockchain platform. It features comprehensive permission management mechanisms and privacy protection policies, making it well suited for enterprise-level applications that demand high data security and privacy preservation. Fabric offers a rich set of SDK interfaces and toolchains, enabling the rapid development and deployment of required smart contracts. This supports functionalities such as CTI information chaining, pricing, and authorization record-keeping. Through these capabilities, Fabric facilitates secure and efficient business processes while ensuring compliance with stringent data-handling requirements.

2.2 CTI Sharing

For an intelligence sharing platform to function effectively, the reward system must be transparent and resistant to manipulation. If the system lacks credibility, it fails to incentivize participation [10]. To address this, blockchain technology, with its immutability and automated execution capabilities via smart contracts, can be leveraged to construct a fair and transparent reward management system [11].

For instance, He *et al.* [5] proposed a blockchain-based threat intelligence sharing and rating system. Their system stores key security information, such as IP addresses, domain names, URLs, security events, vulnerabilities, threat intelligence source credibility, and contribution levels on the blockchain to ensure immutability. Ahmed El-Kosairy *et al.* [7] introduced a consensus algorithm model known as "Proof of Reputation (PoR)" , specifically tailored for CTI sharing and exchange. Their study discusses how blockchain enhances data integrity, non-repudiation, and secure sharing within decentralized networks.

Secure Cyber Threat Intelligence Sharing (SeCTIS) is an innovative framework designed to address the privacy challenges in current information-sharing methods, which often leave organizations vulnerable to data leaks[12]. By integrating swarm learning and blockchain technologies, SeCTIS enables businesses to collaborate while preserving the privacy of their CTI data. Additionally, it incorporates mechanisms to assess both data and model quality, as well as the trustworthiness of participants, through validators utilizing Zero Knowledge Proofs.

The application of blockchain in CTI sharing offers several advantages, such as enhanced data security, transparency, and trust. Blockchain ensures that CTI data is tamper-proof by recording every transaction or data exchange on an immutable ledger, which prevents unauthorized alterations [13]. It also facilitates decentralized control, eliminating single points of failure and enabling collaborative sharing without compromising privacy. Furthermore, the transparency provided by blockchain allows participants to verify the authenticity and origin of CTI data, increasing confidence in shared information [14]. However, there are challenges, including scalability and performance issues. Blockchain, particularly public chains, may struggle to handle the large volume and high speed of data transactions required for CTI sharing in real-time environments. The decentralized nature also means that governance can be complex, requiring careful management of access control and participation rules. Additionally, blockchain's reliance on cryptographic mechanisms can impose computational overheads, impacting efficiency. Thus, while blockchain provides a robust framework for secure and transparent CTI sharing, practical implementation requires careful consideration of its limitations and trade-offs.

3 Design and Implementation

3.1 Plarform Architecture

As illustrated in Fig. 1, the overall framework of the CTI sharing system is built on the Hyperledger Fabric consortium blockchain and the InterPlanetary File System (IPFS). The platform defines three key roles: platform administrators, requesters, and intelligence providers.

Platform administrators are composed of multiple organizations responsible for maintaining the operation of the consortium blockchain, managing on-chain

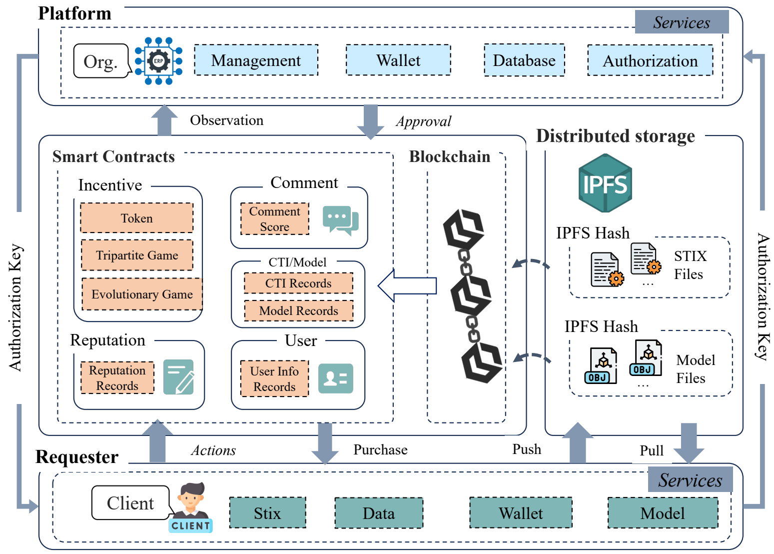


Fig. 1. The overview of the CTI sharing platform architecture.

smart contracts, maintaining the authorization key database, and monitoring on-chain activities, including promptly processing and reviewing requests. All operations are conducted through specialized client terminals. In addition to platform administrators, all other users access the intelligence-sharing platform through their user terminals. The user terminal includes several core modules: a wallet module, an intelligence data conversion module, a model component, and a network module. The wallet module generates a user wallet using asymmetric encryption algorithms (Elliptic Curve secp256k1). First, a public/private key pair is generated locally. The public key creates an on-chain account via the user contract. All on-chain operations require signing with the wallet's private key. After successful verification, the user can interact with smart contracts on the blockchain. The intelligence data conversion module converts local intelligence data, such as traffic and logs, into a standardized threat intelligence format. The model component enables users to train models using their local data or data authorized and downloaded from IPFS. This helps process data that is difficult to standardize.

Requesters are entities in need of threat intelligence, such as schools, enterprises, and government organizations. They gain access to intelligence by purchasing it with platform token points. To earn token points, requester users must also share their intelligence data, meaning they can act as both requesters and intelligence providers.

Intelligence providers are users who supply intelligence to the platform. The role of an intelligence provider is flexible, allowing a user to assume multiple roles, including platform administrator, CTI requester, and intelligence provider. All interactions are conducted through the user terminal.

The CTI-sharing system is built on five essential smart contracts (referred to as chaincode in Fabric): the user contract, intelligence contract, comment contract, incentive contract, and reputation contract.

- `User Contract` manages on-chain user information, user account registration, and user information queries.
- `Intelligence Contract` stores summary information about intelligence (or models), including the intelligence ID, creator, type, IPFS address, and price. It provides functions for querying and purchasing intelligence. Once users purchase intelligence using tokens, they receive an authorization key from the platform to access the encrypted intelligence or model data stored on IPFS.
- `Comment Contract` serves users who purchase intelligence (or models). After using the intelligence or model, users can assess its quality. The Incentive Contract adjusts the intelligence price based on user evaluations, rewarding users with tokens for providing feedback. This incentivizes users to participate in the evaluation process, thus improving the overall quality of shared intelligence.
- `Incentive Contract`: This contract calculates the intelligence price using three pricing algorithms: a basic token-based calculation, a three-party game theory-based algorithm, and an evolutionary game theory-based algorithm (the latter two are part of our prior research). These algorithms share parameters like platform users, intelligence demand, and evaluation scores but differ in computation. The price calculation also incorporates user reputation data, enhancing the robustness of the incentive mechanism.
- `Reputation Contract` tracks users' reputation information, which is integrated into the incentive mechanism. It monitors behaviors such as intelligence sharing, purchasing, and commenting, and uses reputation scores to ensure accurate price calculations and identify malicious users. The reputation algorithms are publicly available and transparent, helping platform administrators address harmful behaviors. This contract is crucial for the stable operation of the intelligence-sharing platform.

The CTI sharing system is built on blockchain technologies, which integrate IPFS for storing the shared standardized threat intelligence (STIX) and model files. Unlike traditional protocols like HTTP, which retrieve data from specific servers, IPFS uses content addressing to identify and access data based on its content. This approach ensures more efficient and resilient data distribution.

In terms of system access, the platform's terminal and the ordinary user's terminal serve distinct purposes. Platform administrators are responsible for managing the entire intelligence-sharing platform and maintaining the consortium blockchain. Consequently, their terminals offer additional functionalities, such as blockchain network management, intelligence authorization, and contract event review (including evaluations of incentives and comments). In contrast, the user terminal provides open interfaces for querying blockchain information, enabling easy system expansion and greater user flexibility.

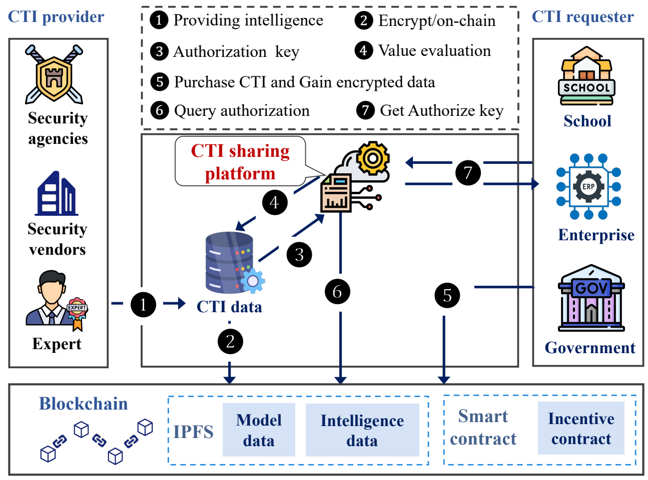


Fig. 2. The workflow of the CTI sharing platform architecture.

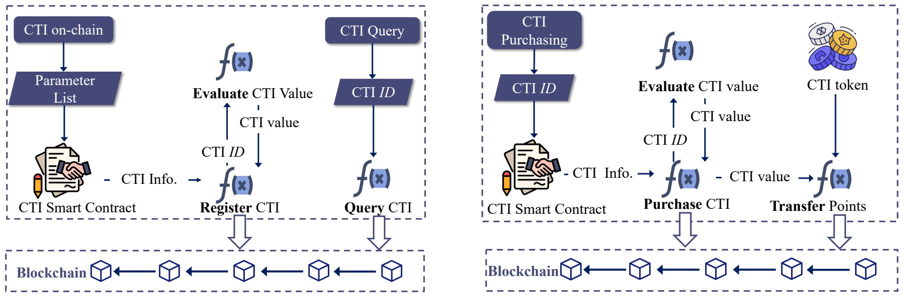
As illustrated in Fig. 2, the workflow of the CTI platform can be explained with seven steps. In step ❶, the CTI provider comprises security agencies, security vendors, and experts providing intelligence to the CTI data center. After the intelligence data is standardized or model training is completed, the STIX data and model data are uploaded to IPFS, while the summary information (such as the IPFS address) is uploaded to Fabric(step ❷). The CTI data center provides an authorization key (❸) to the CTI sharing platform to evaluate the intelligence value (step ❹).

The CTI requester can purchase authorization for CTI information by invoking the on-chain smart contract and then retrieve the encrypted CTI data from IPFS (step ❺). The sharing platform continuously queries the on-chain authorization events in real time and updates the authorization database (step ❻). Afterward, the CTI requester can query the platform and obtain the data authorization key (step ❼). As the data reviewer, the Intelligence Platform securely stores the authorization keys, ensuring safety and reliability. Additionally, by uploading large files to IPFS, the platform reduces the storage load on the federated blockchain, allowing it to scale efficiently and maintain stable operation as the user base grows.

3.2 Smart Contract Desgin

The CTI Smart Contract (CTISmartContract) records the CTI information uploaded to the sharing platform. To minimize storage pressure on the blockchain, only the hash of the raw CTI data is stored on the Fabric blockchain. The key information fields stored on-chain for each CTI entry include CTIID, CTIHash, OpenSource, CreatorUserID, Tags, and StixIPFSHash, among others.

After the CTI data undergoes standardization, the user can choose whether to upload the CTI to the blockchain. The client initiates a smart contract call,



(a) The logic of CTI contract function. (b) The incentive contract function.

Fig. 3. The design principle of contract function.

and the CTI information is uploaded to the Fabric blockchain, while the CTI file itself is uploaded to IPFS. Additionally, the CTI information can be queried on the blockchain using the CTI ID (_ctiId). The function call logic for the CTI on-chain contract is illustrated in Fig. 3(a).

The incentive contract manages user points within the platform, recording point transactions such as when users use points to purchase intelligence authorization. It tracks authorization transactions by transferring points from the buyer's account to the CTI publisher's account. The function call logic of the incentive contract is illustrated in Fig. 3(b), which shows the interaction between the buyer, the contract, and the publisher. The pricing contract evaluates and

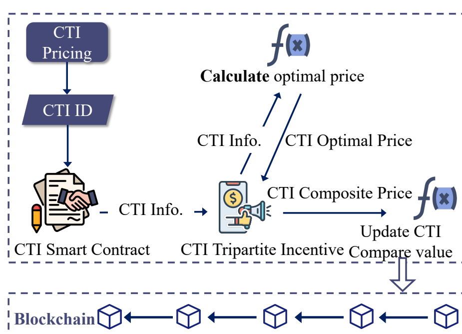


Fig. 4. The pricing contract function.

determines the value of CTI based on the derived game theory model. In this section, we introduce the principle of the tripartite game model used for threat intelligence token pricing. The optimal pricing formula derived from the three-party game theory model is used to design the pricing algorithm [6]. This pricing algorithm takes several parameters as inputs, including the number of CTI publishers, CTI demand, threat level, evaluation confidence, and weight parameters.

These input parameters are treated as independent variables and are used in the pricing contract to calculate the comprehensive value of the CTI. Fig. 4 depicts the pricing workflow. This approach ensures that the pricing system not only responds to market conditions but also incentivizes high-quality intelligence sharing, ensuring the platform's overall sustainability.

3.3 Reputation and Pricing Model

In the CTI sharing platform, to ensure the quality of user-contributed intelligence, the accuracy of the incentive mechanism's point pricing, and to prevent malicious users from disrupting platform operations, we designed a `reputation model` and a `pricing model`. The reputation calculation formula for the `reputation model` is defined as follows.

$$Re_i^t = Re_i^{t-1} + \lambda \left(WR_i - \overline{WR} \right) Re_i^{t-1}, \tag{1}$$

where Re_i^{t-1} represents the reputation score of the user i at the previous time step. The parameter λ is the reputation increment or decay coefficient, controlling the magnitude of reputation score changes. WR_i denotes the user's reputation weight coefficient and WR is the average reputation weight coefficient across all platform users. The difference between a user's reputation weight coefficient and the average reputation weight coefficient determines whether the user's reputation increases or decreases. The user's reputation weight coefficient can be calculated as follows.

$$WR_i = 1 - m_1 * Txd_i + m_2 * Upd_i + m_4 * Ul_i + m_3 * \theta_i, \tag{2}$$

where Txd_i represents the difference between the number of intelligence purchases made by the user i in a week. Upd_i denotes the difference between the number of intelligence uploads by the user i in a week and the average number of intelligence uploads by all platform users during the same period. Ul_i represents the user's level; when a user actively uploads intelligence and earns points exceeding a certain threshold, their level increases. A higher user level indicates greater engagement, which corresponds to a higher reputation weight. Furthermore, θ_i measures the evaluation similarity of the user i, calculated as the average cosine similarity between the user i' evaluation scores and those of other platform users. A higher θ_i indicates that the user's evaluations align closely with the platform's average evaluations, suggesting a higher likelihood of being an active and normal user.

Txd_i is expressed as follows.

$$Txd_i = \begin{cases} Txn_i - \overline{Txn}, & Txn_i > \overline{Txn}, \\ 0, & Txn_i \leq \overline{Txn}, \end{cases} \tag{3}$$

where Txn_i is the number of intelligence purchases by the user i in a week, and $\overline{Txn}$ is the average weekly intelligence purchases across all platform users. If a user's purchase volume significantly exceeds the platform's average threshold, it is considered anomalous, and their reputation weight will be correspondingly reduced.

Upd_i is expressed as follows.

$$Upd_i = \begin{cases} Upn_i - \overline{Upn}, & Upn_i > \overline{Upn}, \\ 0, & Upn_i \leq \overline{Upn}, \end{cases} \tag{4}$$

where Upn_i is the number of intelligence uploads by the user i in a week, and $\overline{Upn}$ is the average weekly intelligence uploads across all platform users. When a user's intelligence upload volume exceeds the platform's average threshold, they are considered an active user, and their reputation weight is correspondingly increased.

when a user i provides a rating for intelligence k at time t, we utilize a user evaluation trust model to compute a composite evaluation score.

$$C_k^t = \eta C_k + (1-\eta) \sum_{i=0}^{m} WC_{i,k} * C_{i,k}, \tag{5}$$

The parameter η is the evaluation weight coefficient, used to adjust the influence proportion of historical evaluations C_k on the composite evaluation score. The term m represents the number of evaluations. The right-hand side of Eq. (5) represents the composite evaluation score for intelligence k.

When user i provides a rating $C_{i,k}$ for intelligence k, the weight of this rating among all ratings for intelligence k is denoted as $WC_{i,k}$.

$$WC_{i,k} = \frac{Re_i PR_i}{\sum_{j=0}^{n} \sum_{k=0}^{m} WC_{j,k}}, \tag{6}$$

The user's reputation score Re_i determines the significance of their evaluations; the higher the reputation score, the greater the weight assigned to them.

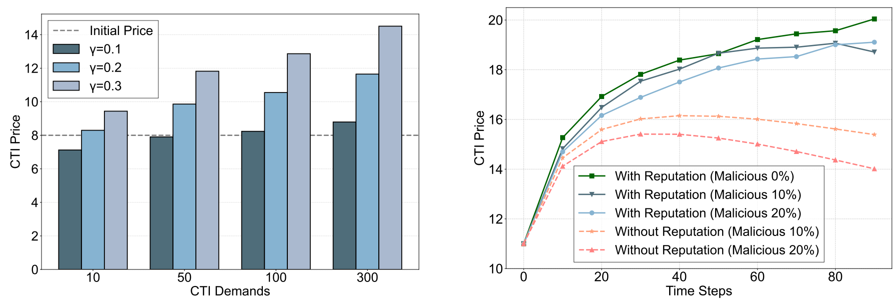
Additionally, PR_i represents the ranking of the user i within the platform's intelligence evaluation graph. This directed graph, denoted as $G(N, V)$, comprises nodes Nrepresenting users and directed edges Vindicating evaluation relationships. To compute the ranking PR_i, we employ the PageRank algorithm, which is defined as follows.

$$PR_i = \frac{1-d}{N} + d \sum_{j \in M(i)} \frac{PR_j}{L(j)}, \tag{7}$$

where N is the total number of users on the platform, d is the damping factor, typically set around 0.85, representing the probability that a user will continue following links, $M(i)$ is the set of users who have evaluated intelligence uploaded by the user i. $L(j)$ is the total number of evaluations made by the user j.

Therefore, when a user utilizes points to purchase intelligence, we employ an incentive model to determine the intelligence's point price. This model integrates the composite evaluation score C_k^t obtained from previous steps to calculate the pricing.

$$\text{price}_k = \alpha * \text{price}_k^{(t-1)} + \beta * \log\left(C_k^t\right) + \gamma \log\left(\text{need}_k\right), \tag{8}$$



(a) CTI price comparison under different demands and demand sensitivities.

(b) CTI price under different malicious user ratios (with reputation vs. without repuation).

Fig. 5. The CTI price evaluation under diverse scenarios.

where price $_k^{(t-1)}$ is the historical point price of intelligence k. price $_k^0$ can be determined using our previously proposed tripartite pricing model and evolutionary game incentive model [6]. C_k^t denotes composite evaluation score at the time t, calculated using Eq. (5). $need_k$ is the historical purchase volume (demand) of intelligence k. α, β, and γ are weight coefficients that control the influence of each factor on the point price calculation, with γ representing demand sensitivity.

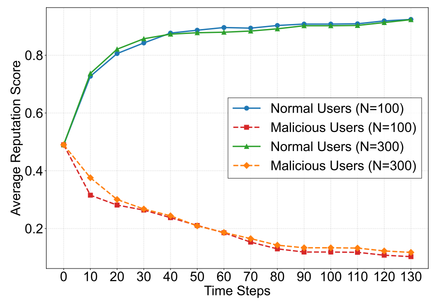
4 Experiments and Analysis

4.1 Environmental Settings

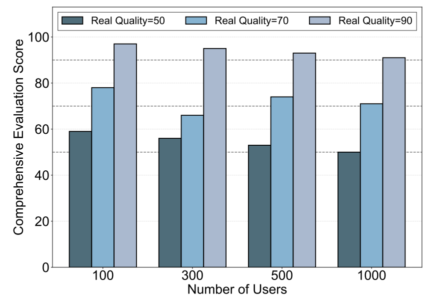
We deployed the CTI sharing platform and blockchain infrastructure on an Ubuntu 22.04.5 server with an Intel(R) Core(TM) i9-13900K CPU and 126 GB of memory. The deployment utilized Hyperledger Fabric version 2.2.0 and IPFS version 0.5.0. Meanwhile, we established a blockchain network comprising three organizations using Fabric, with each component deployed in Docker containers. Furthermore, IPFS was configured to run as a private node on the same server. The user-client applications were developed in Python, which could be deployable on Windows 10 systems. To evaluate the effectiveness of the incentive mechanisms and the designed reputation module, we implemented numerical experiments using Python. We conducted two sets of experiments to assess the impact of user demand and demand sensitivity on intelligence pricing, as well as the effect of the platform user scale on the accuracy of the comprehensive intelligence evaluation score.

4.2 Results and Analysis

As discussed in the previous sections, the intelligence price is primarily influenced by three factors: historical price, evaluation score, and intelligence demand, where intelligence demand refers to the number of purchases. These three



(a) The average reputation score varies with time steps (normal users vs. malicious users).



(b) CTI comprehensive evaluation score under different user numbers and real qualities.

Fig. 6. The testing of average reputation score and CTI comprehensive evaluation score.

factors are weighted by three parameters, α, β, and γ, respectively, to calculate the final comprehensive intelligence price.

In our experiments, the intelligence price is calculated under different demand volume and price sensitivity conditions. The initial price of intelligence is set to 8, with both intelligence quality and evaluation scores set to 70. The resulting intelligence prices under various demand levels and price sensitivities are shown in Fig. 5(a). The experiment shows that with low demand (e.g., 10 units), the intelligence price drops below 8, incentivizing purchases and boosting overall profits. As demand increases, the price rises, reflecting that higher-quality intelligence attracts more buyers. This dynamic encourages the sharing of valuable intelligence and underscores the importance of demand volume and sensitivity in setting prices and fostering the platform's incentive mechanism.

Subsequently, we compared the changes in intelligence prices calculated through the incentive mechanism under different malicious user ratios, both with and without the reputation model, using the price calculated with no malicious users as the baseline. The intelligence prices calculated by the incentive theory model are shown in Fig. 5(b). In this experiment, the malicious user ratios were set at 0%, 10%, and 20%, with the user number fixed at 500. The intelligence price curve in the figure represents intelligence uploaded by a normal user, with a real quality score set to 90. As demonstrated in Fig. 5(b), as the time steps increase, the intelligence prices calculated using the incentive model with reputation are close to the baseline price (no malicious users). However, when the reputation model is absent, the calculated intelligence price is significantly lower than the baseline. This is because malicious users contributed a large number of low scores for this intelligence, which, according to the price calculation formula, results in a decrease in the price due to the lower composite evaluation.

Furthermore, we calculated the average reputation scores of normal and malicious users over time steps, as depicted in Fig. 6(a). It illustrates that as time steps increase, the average reputation of normal users remains stable, while the reputation of malicious users gradually decreases. This proves that our reputation model can effectively detect malicious users on the threat intelligence sharing platform.

Finally, we analyze the accuracy of the comprehensive intelligence evaluation score under different platform user scales. In this experiment, we set platform user counts to 100, 300, 500, and 1000, and evaluated three different intelligence quality levels: low quality (50), medium quality (70), and high quality (90). The comprehensive evaluation scores of the intelligence under different platform user scales and quality levels are shown in Fig. 6(b). In this experiment, we introduced a 10% error margin in the evaluation score given by normal users, relative to the true quality score of the intelligence. This fluctuation is in line with real-world scenarios, where users typically provide reasonably accurate evaluation scores, albeit with a certain degree of variation. The larger the platform user scale, the closer the comprehensive evaluation score is to the true quality score of the intelligence. In the theoretical model section, the comprehensive evaluation score calculates each user's evaluation weight and performs a weighted average of all users' evaluations. As the platform user scale increases, the number of evaluations generated for the intelligence also increases. According to statistical principles, the comprehensive evaluation becomes more accurate and approaches the true evaluation of the intelligence quality.

5 Conclusion

In this paper, we developed a consortium blockchain-based CTI sharing platform using Hyperledger Fabric as the underlying infrastructure, which is empowered by a multi-party game-theoretic model that considers factors like the number of intelligence releases, market demand, and service evaluations to calculate optimal pricing strategies. Through continuous technological innovation and service model upgrades, we aim to foster a collaborative ecosystem where all participants benefit mutually. This approach provides enterprises with superior cybersecurity intelligence sharing solutions and contributes to the advancement of the cybersecurity industry. For future work, we will focus on the scalability and security issues of this CTI sharing platform.

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